

Sales Forecasting & Demand Analysis Report

My Company · April 30, 2026

Executive KPI Summary

Total Revenue \$60,513,867	Avg Monthly Revenue \$1,008,564
CAGR 16.2%	Return Rate 2.85%
Marketing ROI (95% CI) 7.6x [7.4x – 7.8x]	Best Product Product C
Best Region North	Revenue Volatility (CV) 23.1%

1. Historical Sales Performance

Total monthly revenue aggregated across all products and regions. Dashed = 3-month rolling average; dash-dot = centred moving-average trend. Red × markers = rolling Z-score anomalies (threshold 2.5σ).

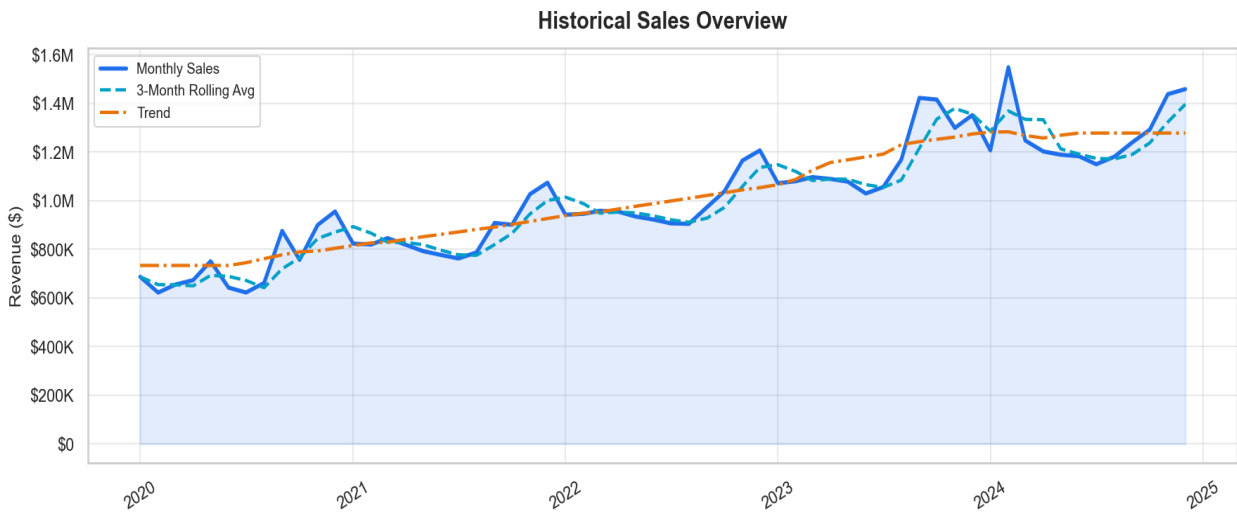


Figure 1: Historical Sales with Trend & Anomalies

Year-over-Year Growth

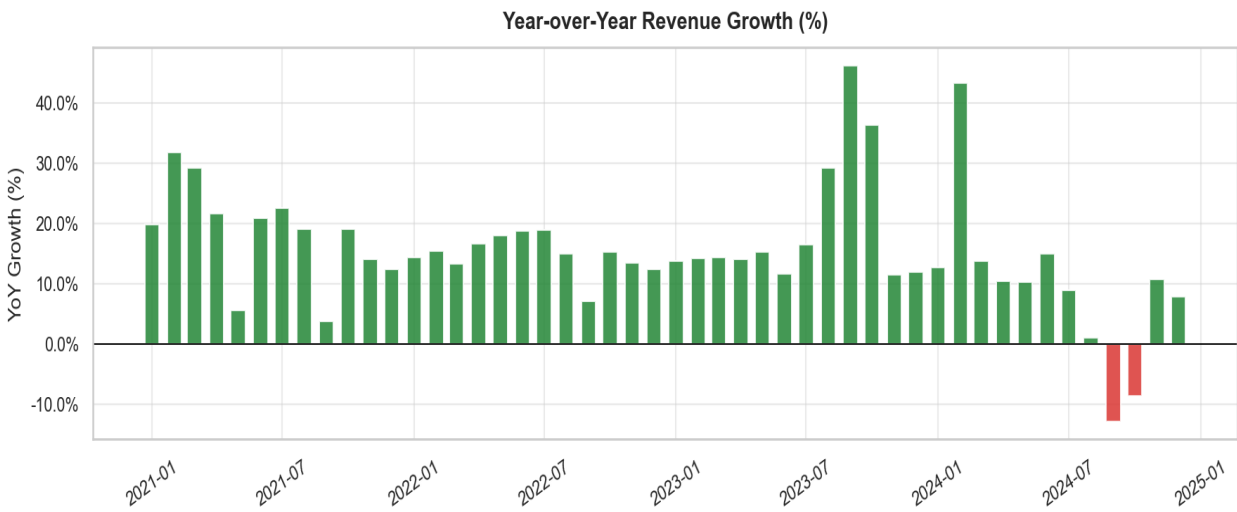


Figure 2: YoY Revenue Growth (%)

Seasonality Heatmap

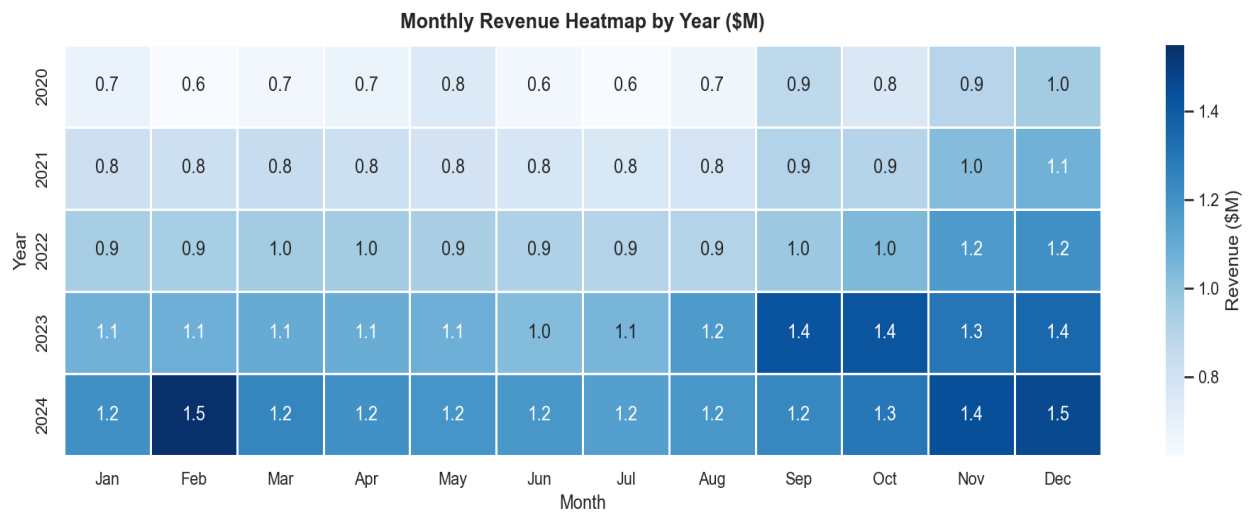


Figure 3: Monthly Revenue Heatmap by Year (\$M)

2. Time Series Decomposition

Series decomposed into Original, Trend (centred 12-period moving average), Seasonality, and Residual noise (additive model). Dominant seasonal period detected: **12 months**.

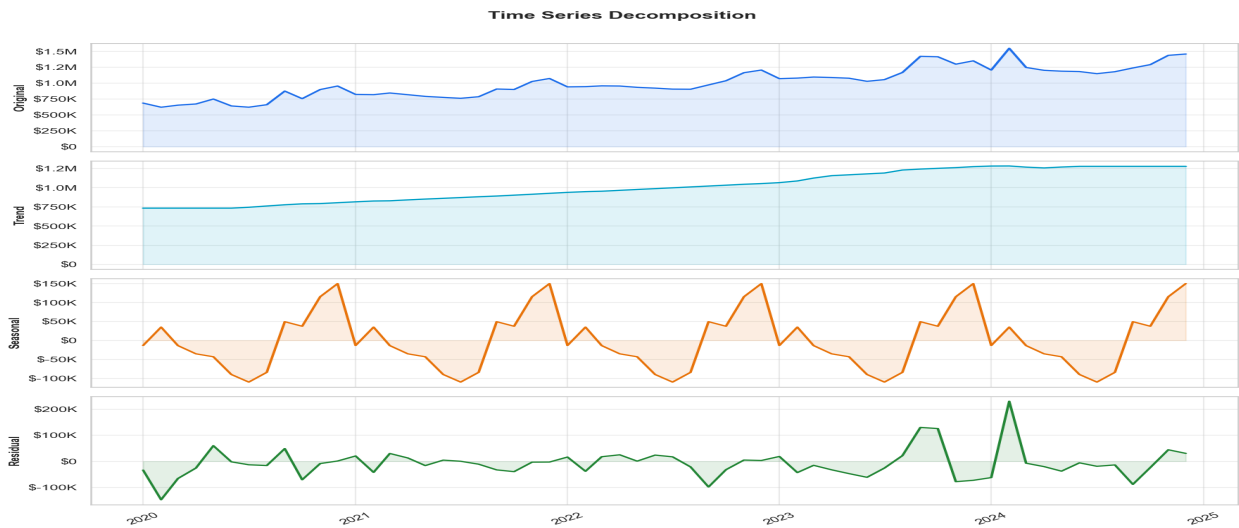


Figure 4: Decomposition — Trend / Seasonal / Residual

3. Product-Level Analysis

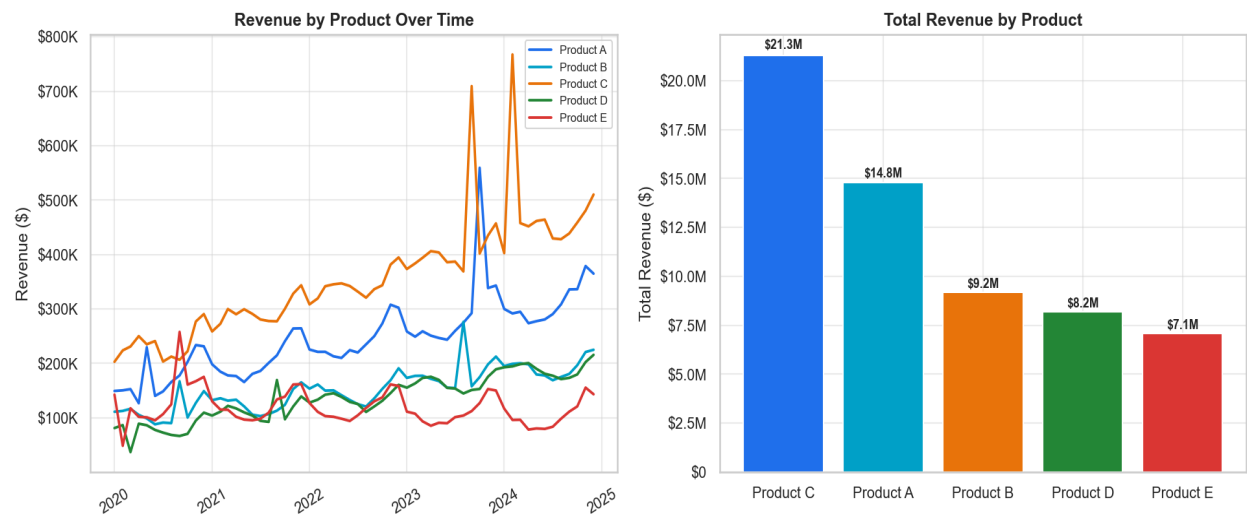


Figure 5: Revenue by Product — Timeline & Total

Product	Total Revenue (\$)	Avg Monthly (\$)	Units Sold
Product C	\$21,285,991	\$354,767	542,548
Product A	\$14,777,263	\$246,288	396,006
Product B	\$9,184,353	\$153,073	231,181
Product D	\$8,172,970	\$136,216	203,237
Product E	\$7,093,291	\$118,222	181,769

4. Demand Forecasting Results

Five models trained and evaluated via walk-forward cross-validation (expanding window, no look-ahead). Stacking ensemble uses inverse CV-sMAPE weights. Best single model by sMAPE: **Holt-Winters**.

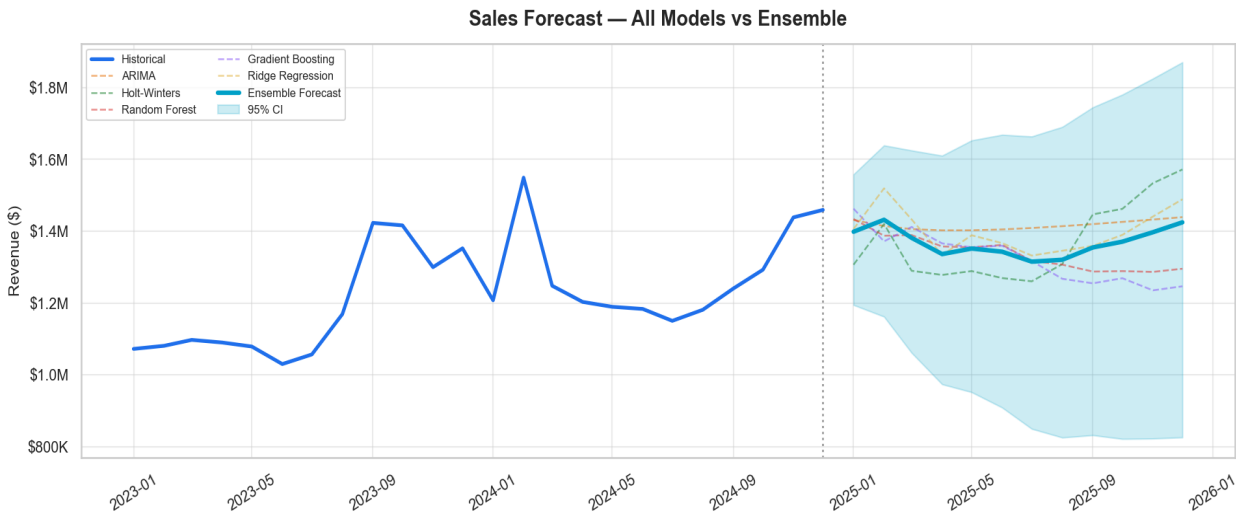


Figure 6: Sales Forecast — Models vs Ensemble (95% CI)

Model Performance Comparison (sMAPE / MASE / RMSE / R²)

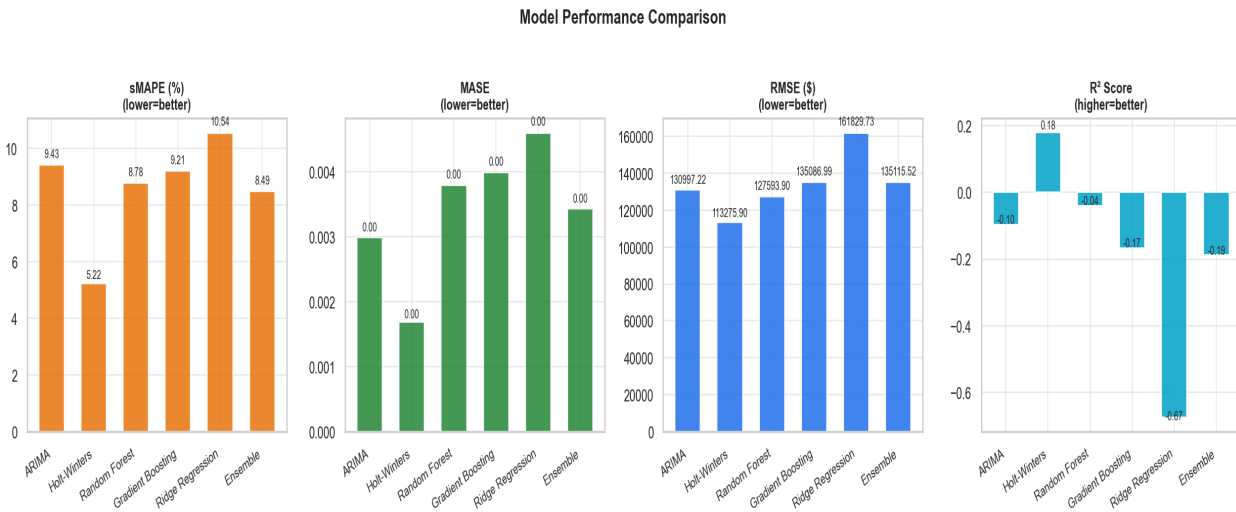


Figure 7: Model Comparison

Model	sMAPE (%)	MASE	RMSE (\$)	R²	CV sMAPE	Time (s)
ARIMA	9.43	0.003	\$130,997	-0.097	15.4	32.4
Holt-Winters	5.22	0.002	\$113,276	0.180	5.6	3.3
Random Forest	8.78	0.004	\$127,594	-0.041	8.4	4.5
Gradient Boosting	9.21	0.004	\$135,087	-0.167	7.5	1.7
Ridge Regression	10.54	0.005	\$161,830	-0.675	5.3	0.2
Ensemble ★	8.49	0.003	\$135,116	-0.189	—	0.0

Residual Diagnostics

Holt-Winters | Ljung-Box $\times$ ($p=0.026$) Shapiro-Wilk $\times$ ($p=0.001$) DW $\checkmark$ (1.68)

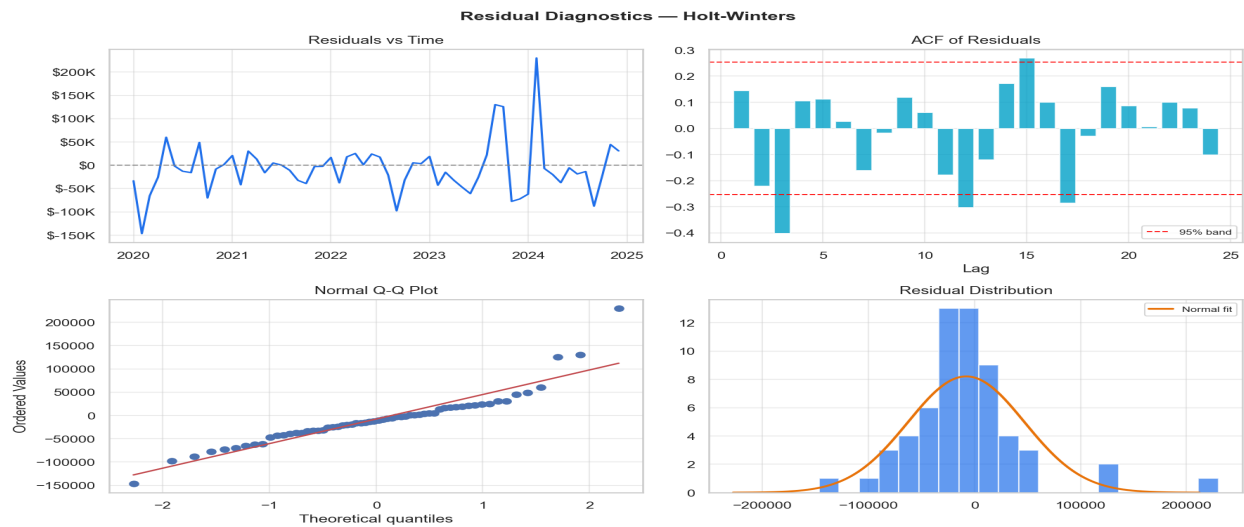


Figure 8: Residual Diagnostics

5. Monthly Forecast Detail

Ensemble monthly point forecast with 95% confidence intervals and month-over-month change.

Month	Forecast (\$)	Lower CI (\$)	Upper CI (\$)	CI Width (\$)	MoM Δ
Jan 2025	\$1,398,350	\$1,194,464	\$1,558,017	\$363,553	-4.2%
Feb 2025	\$1,431,676	\$1,161,988	\$1,638,828	\$476,840	+2.4%
Mar 2025	\$1,381,289	\$1,061,283	\$1,624,705	\$563,423	-3.5%
Apr 2025	\$1,336,110	\$973,659	\$1,610,124	\$636,466	-3.3%
May 2025	\$1,351,554	\$951,586	\$1,652,644	\$701,058	+1.2%
Jun 2025	\$1,342,662	\$908,624	\$1,668,386	\$759,762	-0.7%
Jul 2025	\$1,315,054	\$849,524	\$1,663,592	\$814,068	-2.1%
Aug 2025	\$1,320,268	\$825,269	\$1,690,196	\$864,927	+0.4%
Sep 2025	\$1,354,607	\$831,783	\$1,744,774	\$912,991	+2.6%
Oct 2025	\$1,370,681	\$821,401	\$1,780,128	\$958,727	+1.2%
Nov 2025	\$1,396,960	\$822,387	\$1,824,876	\$1,002,489	+1.9%
Dec 2025	\$1,424,576	\$825,713	\$1,870,260	\$1,044,546	+2.0%

6. Strategic Insights & Recommendations

- 1. Revenue Outlook: \$16,423,788 forecast over the next 12 months (+35.7% vs trailing same-period avg).
- 2. Top Performer: **Product C** leads all products. Consider increasing inventory allocation and marketing budget.
- 3. Seasonal Strategy: 12-month seasonality detected. Begin inventory build 6–8 weeks ahead of Q4 peak.
- 4. Anomaly Management: **0** anomalies flagged — review for bulk orders, promotions, or data-entry errors.
- 5. Model Confidence: Best single model sMAPE = **5.2%** (MASE=0.00) | Ensemble sMAPE = **8.5%**. CV sMAPE (walk-forward) provides unbiased out-of-sample error estimate.
- 6. Marketing ROI: **7.6x** (95% bootstrap CI: 7.4x – 7.8x). Interpret the full interval, not just the point estimate.
- 7. Returns: Return rate is **2.85%**. Review high-return SKUs to improve net revenue materially.

Auto-generated by Sales Forecasting System. ARIMA: exact MLE via innovations form, ψ -weight CI propagation (dead_undifference removed). Holt-Winters: SSE minimisation, AIC variant selection, analytical/bootstrap CI. ML models: PACF-guided features, feature matrix built once per run, iterative multi-step forecast, bootstrap CI. Ensemble: inverse CV-sMAPE stacking (walk-forward, non-leaky) with NaN guard. sMAPE symmetric and bounded [0,200]; MASE scaled to naïve baseline. Marketing ROI includes 95% bootstrap CI. Charts dir: system tempdir (cross-platform). Forecasts carry inherent uncertainty — combine with domain expertise.